

7 Potassium short overview

Potassium supports water regulation and crop function, but a low leaf value does not automatically prove inadequate fertilizer supply.

Why potassium matters

- **Water regulation.** Potassium helps control stomata and cell water balance.
- **Enzyme activity.** It activates many processes involved in growth and energy use.
- **Transport.** It helps move sugars and other products of photosynthesis.
- **Crop demand.** Fruit and strong vegetative growth can increase demand and change seasonal leaf values.

What controls potassium availability

Factor	Likely effect on K
Soil solution and exchange sites	Roots take up dissolved K while exchange sites on clay and organic matter replenish the solution.
Texture and clay type	Sandy low-CEC soils hold less K; some clays can temporarily fix K between mineral layers.
Organic matter	Adds cation-holding sites and can improve rooting water supply and nutrient cycling.
pH	Usually an indirect control through root health exchange chemistry and competition rather than simple K precipitation.
Water supply	Dry soil reduces diffusion and root uptake; saturated soil reduces oxygen and root function.
Salinity and sodium	Salt stress reduces water uptake; high sodium can compete and can damage infiltration.
Other cations	Very high magnesium calcium ammonium or potassium can shift competition and tissue balance.
Roots and crop load	Weak roots restrict access; heavy fruit demand can draw down mobile K in leaves.

What leaf results can and cannot say

- **Tentative UC guide.** For avocado leaves, K below 0.35 percent is listed as deficient, 0.75 to 2.0 percent as adequate, and above 3.0 percent as excess.
- **Use ranges as guides.** Cultivar, leaf age, sampling date, crop load, rootstock, water status, and laboratory method can affect interpretation.
- **Low leaf K can reflect several causes.** Low soil supply is one possibility, along with dry or damaged roots, poor distribution, salinity, heavy crop demand, or competition.
- **High leaf K is not automatically beneficial.** Unneeded application can raise salt load and alter calcium or magnesium balance.

8 Potassium diagnostic sheet

Confirm the limiting process before choosing a source, rate, placement, or timing.

Field checks in recommended order

Check	Question
Sampling quality	Were comparable mature leaves collected from the correct position and season?
Pattern	Is the issue tied to blocks soil types emitters crop load or root disease areas?
Water salts and irrigation	What are water EC sodium chloride and bicarbonate and are flow distribution and wetted volume adequate?
Soil K and CEC	What are solution plus exchangeable K texture clay type organic matter and CEC in the wetted root zone?
Competing cations	Are calcium magnesium sodium or ammonium unusually high relative to the setting?
Root condition	Are feeder roots abundant light colored and active or sparse black brittle and waterlogged?
Demand and history	Was crop load or growth unusually high and what K sources rates and responses were recorded?

Possible responses after diagnosis

- **If soil supply is truly low.** Select a rate and placement that match soil holding capacity, root distribution, crop demand, and the salinity budget.
- **If roots are dry wet or diseased.** Correct irrigation drainage and root health before expecting fertilizer alone to restore uptake.
- **If salinity or sodium is high.** Address water source, uniformity, leaching feasibility, drainage, and ion balance. Avoid adding unnecessary salts.
- **If K is adequate.** Do not treat a model association or visual symptom as proof of need.

Phase 1 link

Potassium and chloride appeared together in important model interactions. The practical meaning is to inspect nutrient supply and salt exposure together. It is not proof that potassium application will correct chloride injury.

Key potassium cautions

- Count the accompanying ion. Potassium chloride adds chloride and is a poor fit where chloride exposure is already a concern.
- A tissue increase shows uptake but does not by itself prove a profitable yield or quality response.
- Compare K with calcium magnesium sodium chloride salinity and roots rather than chasing one ratio.